

Final Comparison and Consolidated Winning Blueprint

1. High-level verdict

All three documents are strongly aligned on the same winning strategy:

Do **not** build a large multi-model research system.
Build a small, working, prevention-first prototype where voice risk evidence blocks or forces independent verification before a sensitive action can complete.

Each document has a different strength:

Document	Strength	Best use
ChatGPT.pdf	Judge-facing product narrative, MVP simplification, proof package	Use for PPT storytelling, demo framing, and judge Q&A
Qwen.pdf	Implementation-ready deliverables: PRD, DB schema, API, worker pipeline, infra	Use for actual build and technical appendix
Claude.docx	Architecture discipline, validation gates, earned branches, security boundaries	Use for credibility, roadmap, and responsible claims

The best final approach is:

ChatGPT's judge narrative + Qwen's engineering implementation + Claude's validation and governance gates.

2. Ideas all three documents align on

These are the core ideas you should treat as final.

2.1 Prevention is the real product, not detection

All three agree:

- Detecting synthetic voice is not enough.
- The system must prevent a sensitive action from completing.
- The winning demo is not “we detected a fake voice.”

- The winning demo is:

A cloned voice requested a sensitive action, the system raised risk, and the action was blocked or forced into independent verification.

Final claim to use:

“Our system converts voice-risk evidence into an authorization decision before a sensitive action completes.”

2.2 Build a small MVP first

All three explicitly say:

- Do not add many AI models.
- Do not build speaker verification, replay model, OOD model, watermarking, and few-shot adaptation in the first prototype.
- Use one detector first.
- Add branches only after the baseline proves value.

Final MVP:

```
Audio stream
-> Silero VAD
-> Quality check
-> One spoof detector
-> Temporal smoothing
-> Deterministic policy
-> Action blocking
-> Independent OTP verification
-> Audit log
```

2.3 Use Silero VAD correctly

All three align on this, with slightly different wording:

- Silero VAD is for speech segmentation.
- It is not a replacement for transport.
- For MVP, avoid full WebRTC complexity.
- Use simple PCM/WebSocket or Web Audio streaming.

Final decision:

Use **Web Audio API / AudioWorklet** -> **WebSocket PCM**
-> **Silero VAD** for the MVP.
Keep WebRTC as a future adapter, not the hackathon critical path.

2.4 Use only one spoof detector in the critical path

All three agree:

- One detector is enough for MVP.
- Preferred detector: frozen SSL representation plus lightweight head.
- Examples: WavLM-base, wav2vec2-base, or similar.
- AASIST/RawNet/LFCC-LCNN can be baseline/fallback/teacher, not the main complexity burden.

Final model strategy:

Priority	Model choice	When
Primary	Frozen WavLM/wav2vec2 + lightweight calibrated head	If GPU or acceptable latency
Fallback	LFCC/LCNN or lightweight RawNet-style model	If CPU-only or latency issue
Later	AASIST, replay branch, ASV branch, OOD branch	Only after MVP works

2.5 Do not output binary “fake/real”

All three reject a simple fake/real label.

Use risk states and bounded language.

Do not say:

“The caller is trusted.”

Say:

“No strong synthetic-speech evidence detected.”

Final risk states:

Risk State	Meaning
LOW	No strong synthetic-speech evidence detected
ELEVATED	Elevated synthetic-speech indicators
HIGH	Strong synthetic-speech indicators
INSUFFICIENT_EVIDENCE	Not enough voiced evidence
POOR_QUALITY	Audio quality too low for reliable evidence
SERVICE_UNAVAILABLE	Detector/stream unavailable

Risk State	Meaning
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2.6 Sensitive actions must require independent verification

All three agree:

- Voice risk should not be the only authentication factor.
- Even LOW risk does not prove identity.
- HIGH/ELEVATED/INSUFFICIENT risk must trigger step-up verification.
- Verification must be independent from the voice channel.
- For demo, simulated OTP is enough.

Final prevention rule:

A sensitive action cannot complete unless the backend verifies a valid, single-use, action-bound confirmation token.

2.7 Fail closed, not fail open

All three agree:

If any of these happen:

- Poor audio
- No speech evidence
- Detector unavailable
- Stream disconnected
- OTP expired
- OTP incorrect
- Inference timeout

Then:

The sensitive action must remain unapproved.

This is a major winning point.

2.8 Indian-language evaluation is mandatory

All three agree that the system must not be English-only.

Minimum language coverage:

1. Hindi
2. Indian English
3. Hinglish
4. One regional language selected by the team

Examples:

- Bengali
- Tamil
- Telugu
- Marathi
- Gujarati
- Kannada

Data sources:

Purpose	Source
Synthetic Indian speech	IndicSynth
Genuine Indian speech	IndicVoices, public datasets, team-collected consented samples
Anti-spoof benchmark	ASVspoof 5
RTC conditions	RTCFake if accessible
Replay	ReplayDF if license permits

Important:

IndicSynth gives synthetic speech. Genuine speech must be sourced separately.

2.9 Short-audio assumption must be validated

All three reject blindly using 0.5-1.5 second windows.

Final requirement:

Run a duration study:

Duration	Purpose
0.5s	Very early alert, likely unreliable
1.0s	Early alert
2.0s	MVP default candidate
4.0s	More stable evidence

Important rule:

Do not repeat or pad a short clip to fake longer evidence.

2.10 Channel, codec, replay, and unseen-generator testing are required

All three agree that benchmark-only performance is weak.

Minimum test conditions:

- Clean audio
- Telephone audio
- Codec degradation
- Noise
- Hinglish/code-switching
- Regional accents
- Replay/re-recording
- Unseen generator family
- Partial manipulation inside a genuine call

This is what makes the submission credible.

2.11 Privacy by default

All three agree:

- Do not store raw audio by default.
- Store only scores, states, reason codes, metadata, and audit events.
- If debug audio is stored, keep short retention.
- Show deletion/retention behavior.
- Do not claim legal compliance certification.

Final privacy statement:

“The prototype demonstrates privacy-preserving design through short-lived buffers, minimal retention, and auditable decision events.”

2.12 Auditability is mandatory

All three agree every decision should be explainable.

Every risk/action event should include:

- Timestamp

- Session ID
- Risk state
- Reason codes
- Evidence duration
- Model version
- Threshold profile
- Action status
- Verification status
- Final outcome

This is very important for judges.

2.13 Local fallback is mandatory

All three agree the demo must not depend on external services.

Fallback requirements:

- Local pre-recorded clips
 - Local inference
 - Local OTP simulation
 - File-upload mode if live mic fails
 - Recorded backup video if live demo fails
-

2.14 Claims must be disciplined

All three agree: avoid overclaiming.

Do not say:

- “We detect all synthetic voices.”
- “We are production-ready.”
- “We outperform all commercial systems.”
- “Green means trusted.”
- “Voice risk equals identity verification.”
- “Watermarking solves the problem.”

Say instead:

- “We reduce risk by forcing verification when evidence is weak or suspicious.”
 - “The system fails closed under uncertainty.”
 - “We report performance by language, duration, and channel.”
 - “Detection is not prevention; authorization control is prevention.”
-

3. Fallback ideas the documents rely on

These are contingency ideas. You should keep them ready.

3.1 If live microphone/streaming fails

Fallback:

- Use uploaded audio files.
- Use pre-recorded attack/genuine clips.
- Use file-upload evaluation endpoint.

What to say:

“The same pipeline works on live streams and recorded files. For reliability, we can demonstrate the recorded path.”

3.2 If WebRTC is too complex

Fallback:

- Do not use WebRTC.
- Use browser Web Audio/AudioWorklet.
- Send PCM over WebSocket.

What to say:

“For the prototype, we use a deterministic PCM/WebSocket stream. WebRTC can be added later as a transport adapter.”

3.3 If GPU is unavailable

Fallback:

- Use a lightweight detector.
- Use LFCC/LCNN or small RawNet-style model.
- Or use a distilled/quantized model.

What to say:

“The architecture supports swappable detectors. We selected the smallest model that meets our latency and evidence requirements.”

3.4 If the detector accuracy is not strong

Fallback:

- Lean on the prevention workflow.
- Show uncertainty handling.
- Show step-up verification.
- Show fail-closed behavior.

What to say:

“Even when detection is uncertain, the system prevents unsafe authorization by requiring independent verification.”

This is actually a strong answer.

3.5 If real OTP/SMS is unavailable

Fallback:

- Simulated OTP app.
- Local OTP console.
- Push/email simulation.

What to say:

“The MVP demonstrates independent verification using a simulated channel. The architecture is channel-agnostic.”

3.6 If PostgreSQL/Redis setup becomes heavy

Fallback:

- Use SQLite for demo.
- Use in-memory state.
- Use JSON/CSV audit log.

But for final submission, PostgreSQL is cleaner.

3.7 If real RTC data is unavailable

Fallback:

- Simulate telephone audio.
- Apply codec/noise augmentation.
- Use 8 kHz downsampling.
- Use Opus/AMR simulation.

What to say:

“We simulate realistic communication degradation and test the detector under those conditions.”

3.8 If IndicSynth or large datasets are not accessible

Fallback:

- Use team-collected genuine clips.
- Use permitted TTS samples.
- Use public licensed Indian speech datasets.
- Keep dataset manifest and consent clear.

What to say:

“We use a curated evaluation manifest with genuine and synthetic samples, and we document source/license limitations.”

3.9 If calibration is incomplete

Fallback:

- Do not call raw scores probabilities.
- Use thresholded risk bands.
- Say calibration is in progress.

What to say:

“The MVP uses thresholded risk states. Calibration is part of the validation roadmap.”

3.10 If advanced branches are not built

Fallback:

- Show them as roadmap.
- Show validation gates.
- Show why they were deferred.

What to say:

“We intentionally avoided adding models that have not yet earned their latency and robustness cost.”

This is a strong engineering answer.

4. Where the documents differ and what final decision to take

4.1 Transport layer

Document	Position
ChatGPT.pdf	Web Audio API/AudioWorklet + WebSocket PCM
Qwen.pdf	Use Silero and lightweight WebSocket/PCM instead of full WebRTC
Claude.docx	WebRTC/PCM is transport; Silero is segmentation; do not confuse them

Final decision

Use:

Browser Web Audio / AudioWorklet
-> WebSocket PCM
-> Server-side Silero VAD

Do not build full WebRTC for MVP.

Say:

“WebRTC is a future transport adapter. For the prototype, we use deterministic PCM streaming over WebSocket.”

4.2 Risk-state naming

Document	Risk/action states
ChatGPT.pdf	NORMAL, STEP-UP, BLOCK, REVIEW
Qwen.pdf	LOW, ELEVATED, HIGH, INSUFFICIENT_EVIDENCE, POOR_QUALITY, SERVICE_UNAVAILABLE
Claude.docx	LOW, MEDIUM, HIGH, AMBIGUOUS; action decisions ALLOW, STEP_UP, BLOCK, REVIEW

Final decision

Separate **evidence state** and **action decision**.

Evidence states:

LOW
ELEVATED
HIGH
INSUFFICIENT_EVIDENCE
POOR_QUALITY
SERVICE_UNAVAILABLE

Action decisions:

CONTINUE_MONITORING
STEP_UP_VERIFICATION
BLOCK_PENDING_REVIEW
REQUIRE_MANUAL_REVIEW

This is the cleanest model.

4.3 API style

Document	API path style
Qwen.pdf	/v1/...
Claude.docx	/api/v1/...
ChatGPT.pdf	Less explicit

Final decision

Use:

/api/v1

It is clean and standard.

4.4 Database depth

Document	DB approach
Qwen.pdf	Full SQL schema with model versions, thresholds, sessions, scores, actions, OTP, audit, eval

Document	DB approach
Claude.docx	Simpler schema with calls, evidence windows, risk states, actions, confirmations, audit
ChatGPT.pdf	Deliverable-oriented, not schema-heavy

Final decision

Use Qwen's schema as the implementation base, but simplify for demo.

Minimum tables:

```

sessions
audio_segments
risk_scores
risk_states
protected_actions
verification_challenges
approval_tokens
audit_events
model_versions
threshold_profiles

```

Optional evaluation tables:

```

eval_runs
eval_results

```

4.5 Worker pipeline

Document	Worker approach
Qwen.pdf	Stream, inference, evaluation, cleanup, metrics
Claude.docx	Capture, inference, policy, confirmation, audit, evaluation, escalation
ChatGPT.pdf	Not as detailed

Final decision

Use this combined worker model:

Worker	MVP required?
Stream worker	Yes
Inference worker	Yes
Policy worker	Yes
Verification worker	Yes
Audit worker	Yes
Evaluation worker	Yes, for reports
Cleanup worker	Optional but useful
Escalation worker	Phase 2

4.6 TrustGraph

Document	TrustGraph position
ChatGPT.pdf	Mentions prevention-first and evidence, less emphasis on TrustGraph
Qwen.pdf	Avoids TrustGraph complexity in MVP
Claude.docx	TrustGraph is a fusion architecture, not a model count; branches earned later

Final decision

Use this wording:

“TrustGraph is the long-term evidence-fusion architecture. The MVP implements only the first branch: one spoof detector plus policy control. Additional branches are earned through validation.”

Do not build 15 models.

4.7 Data sources

Document	Data emphasis
ChatGPT.pdf	Most explicit: IndicVoices, IndicSynth, ASVspoof 5, RTCFake, ReplayDF
Qwen.pdf	IndicSynth + genuine/public/team-collected data

Document	Data emphasis
Claude.docx	IndicSynth + genuine speech separately

Final decision

Use a layered dataset plan:

Layer	Use
IndicSynth	Synthetic Indian-language attacks
IndicVoices/public datasets/team recordings	Genuine Indian speech
ASVspoof 5	General anti-spoof benchmark discipline
RTCFake	RTC/channel evaluation if accessible
ReplayDF	Replay evaluation if license permits

For demo, use a small curated subset.

4.8 Tone and focus

Document	Tone
ChatGPT.pdf	Product, judge proof, competitive narrative
Qwen.pdf	Build specification and implementation artifacts
Claude.docx	Governance, gates, responsible deployment

Final decision

Your PPT should combine all three:

1. Start with ChatGPT's narrative.
2. Show Qwen's technical deliverables.
3. Close with Claude's validation and responsible deployment discipline.

5. Final consolidated blueprint

Use this as your master build and PPT blueprint.

A. Final product identity

Product name

VIGILVOICE - Voice Trust Gate

One-line positioning

A voice-fraud verification layer for Indian-language calls that measures synthetic-speech risk and requires independent confirmation before sensitive approvals.

Core thesis

Existing detectors can fail under unseen generators, codecs, replay, noise, and short audio. Therefore, we do not trust a single fake/real score. We build a calibrated evidence-to-action layer that blocks or pauses sensitive actions until independent verification succeeds.

Winning claim

“Our prototype proves prevention, not just detection.”

B. Final MVP scope

Must build

Component	Required
Browser/local audio capture	Yes
WebSocket PCM streaming	Yes
Silero VAD	Yes
Quality gate	Yes
One spoof detector	Yes
Temporal smoothing	Yes
Deterministic policy engine	Yes
Protected action API	Yes
OTP verification	Yes
Backend action blocking	Yes
Audit log	Yes
One-screen dashboard	Yes
Local fallback clips	Yes
Evaluation report	Yes

Must not build in MVP

Component	Status
Speaker verification	Phase 2
Replay-specific model	Phase 2
OOD model	Phase 2
Few-shot adaptation	Phase 3
Watermarking	Not MVP
Blockchain/provenance	Not MVP
Multi-model ensemble	Not MVP
Full telecom integration	Not MVP
Production multi-tenant SaaS	Not MVP

C. Final system architecture

Final architecture flow

```
Browser / Demo Client
  v
Web Audio API / AudioWorklet
  v
WebSocket PCM Stream
  v
FastAPI Stream Gateway
  v
Silero VAD
  v
Quality Gate
  v
Evidence Window Builder
  v
One Spoof Detector
  v
Calibration
  v
Temporal Smoothing
  v
Policy Engine
  v
Risk State + Reason Codes
  v
Protected Action Control
  v
```

Independent OTP Verification
v
Approve / Block / Review
v
Audit Log

Component table

Layer	Component	MVP implementation	Output
Capture	Audio client	Web Audio/AudioWorklet or file upload	PCM chunks
Transport	Stream gateway	FastAPI	Timestamped audio
Segmentation	VAD	WebSocket Silero VAD	Speech/silence segments
Quality	Quality gate	RMS, SNR, clipping, duration	Quality state
Evidence	Window builder	2s default, validate 0.5/1/2/4s	Voiced evidence
Detection	Spoof detector	Frozen SSL + lightweight head	Spoof score
Calibration	Score mapping	Temperature/Platt or threshold	Calibrated score/risk
Aggregation	Temporal smoothing	EMA/median + persistence	Stable risk
Policy	Decision engine	Deterministic rules	Action decision
Prevention	Action control	Backend blocking	Approved/blocked
Verification	OTP service	Simulated OTP	Confirmed/expired
Audit	Event log	Postgres/JSON	Traceable events

D. Final risk and action model

Evidence risk states

State	UI label	Meaning
LOW	No strong synthetic-speech evidence detected	Conversation may continue
ELEVATED	Elevated synthetic-speech indicators	Step-up verification required
HIGH	Strong synthetic-speech indicators	Block pending review
INSUFFICIENT_EVIDENCE	Insufficient voice evidence	Verification/review required
POOR_QUALITY	Audio quality too low	Verification/review required
SERVICE_UNAVAILABLE	Detection unavailable	Fail closed

Action decisions

Risk State	Sensitive Action Outcome
LOW	May continue, but high-value actions may still require normal app authentication
ELEVATED	Requires independent verification
HIGH	Blocked pending review
INSUFFICIENT_EVIDENCE	Requires verification or review
POOR_QUALITY	Requires verification or review
SERVICE_UNAVAILABLE	Requires verification or review

Important wording

Never say:

“Caller is trusted.”

Say:

“No strong synthetic-speech evidence detected.”

E. Final AI pipeline

Audio input

Property	Value
Sample rate	16 kHz preferred
Channels	Mono
Bit depth	16-bit PCM
Chunk size	100-250 ms
VAD	Silero VAD
Evidence window	2s default
Duration study	0.5s, 1s, 2s, 4s

Silero VAD config

```
vad:
  model: silero_vad
  threshold: 0.5
  min_speech_ms: 250
  min_silence_ms: 500
  speech_pad_ms: 100
```

Quality checks

Check	Purpose
RMS	Too quiet
Clipping	Distorted audio
SNR estimate	Noisy audio
Silence ratio	Insufficient voiced content
Bandwidth	Telephone/narrowband detection
Duration	Avoid short unreliable evidence

Detector selection

Primary:

Frozen WavLM/wav2vec2 encoder + lightweight classifier head

Fallback:

LFCC/LCNN or lightweight RawNet-style detector

Temporal smoothing

Example:

$\text{smoothed_score} = 0.3 * \text{current_score} + 0.7 * \text{previous_smoothed_score}$

State transition rule:

Require N consecutive windows or T milliseconds persistence before changing state.

Example:

```
policy:
  low_max_score: 0.25
  high_min_score: 0.70
  state_persistence_windows: 2
  min_evidence_ms: 2000
```

F. Final worker pipeline

MVP workers

Worker	Responsibility
Stream worker	Receives audio chunks, maintains buffer
VAD/quality worker	Runs Silero VAD and quality checks
Inference worker	Runs spoof detector
Policy worker	Converts scores to risk/action
Verification worker	Handles OTP challenge and approval token
Audit worker	Writes events
Evaluation worker	Runs test matrix and produces metrics
Cleanup worker	Expires OTPs, deletes debug audio

Fail-safe worker rules

Failure	Behavior
Detector timeout	Mark SERVICE_UNAVAILABLE
Poor audio	Mark POOR_QUALITY
No speech	Mark INSUFFICIENT_EVIDENCE
Stream lost	Action remains unapproved
OTP expired	Action remains unapproved
Inference crash	Fail closed

G. Final database schema

Use PostgreSQL for clean demo.

Minimum tables

```
model_versions
threshold_profiles
sessions
audio_segments
risk_scores
risk_states
protected_actions
verification_challenges
approval_tokens
audit_events
eval_runs
eval_results
```

Most important table relationships

```
sessions
|---- audio_segments
|---- risk_scores
|---- risk_states
|---- protected_actions
|      |---- verification_challenges
|      |---- approval_tokens
|      -- audit_events
-- audit_events
```

Retention policy

Data	Default
Raw audio	Not stored
Debug audio	24 hours if enabled
Risk scores	Stored for evaluation
Audit events	Stored for demo/evaluation
OTP	Hashed only
Approval token	Hashed, single-use

H. Final API specification

Use FastAPI.

Base path:

/api/v1

Core endpoints

Method	Endpoint	Purpose
GET	/api/v1/health	Health check
POST	/api/v1/sessions	Create session
GET	/api/v1/sessions/{session_id}	Get session risk summary
WebSocket	/api/v1/stream/{session_id}	Audio stream
POST	/api/v1/actions	Create protected action
GET	/api/v1/actions/{action_id}	Get action status
POST	/api/v1/actions/{action_id}/verification	Request OTP
POST	/api/v1/verifications/{challenge_id}/confirm	Verify OTP
POST	/api/v1/actions/{action_id}/complete	Complete action
GET	/api/v1/actions/{action_id}/audit	Get audit trail
POST	/api/v1/eval/files	Upload audio for offline evaluation

Example risk update

```
{
  "type": "risk_update",
  "session_id": "a1b2c3",
  "timestamp": "2026-09-09T10:02:11Z",
  "risk_state": "ELEVATED",
  "smoothed_score": 0.61,
  "calibrated_score": 0.58,
  "evidence_duration_ms": 2300,
  "latest_evidence_age_ms": 700,
  "reason_codes": [
    "SYNTH_SCORE_ELEVATED"
  ],
  "action_impact": {
    "sensitive_action_allowed": false,
    "requires_verification": true
  }
}
```

I. Final infrastructure plan

Local demo stack

Component	Technology
Frontend	React/Vite or simple HTML/JS
Backend API	FastAPI
Streaming	WebSocket
VAD	Silero VAD ONNX
Model runtime	PyTorch or ONNX Runtime
Database	PostgreSQL
Cache/queue	Optional Redis
Containerization	Docker + Docker Compose
Monitoring	Prometheus/Grafana optional

Hardware

Minimum:

8-core CPU
16 GB RAM
50 GB storage
GPU optional

Preferred:

12+ core CPU
32 GB RAM
NVIDIA GPU with 8 GB+ VRAM

Fallback

Must have:

- Local model.
- Local clips.
- Local OTP.
- File upload.
- No dependency on internet APIs.

J. Final security and privacy model

Threat model

Threat	Mitigation
Synthetic voice impersonation	Risk scoring + independent verification

Threat	Mitigation
Replay attack	Evaluation condition; quality/channel signals; later replay branch
Short-audio evasion	Evidence duration rules
Codec/noise evasion	Channel evaluation and fail-safe policy
Unseen generator	Uncertainty handling and step-up
API abuse	Auth, rate limiting, RBAC
OTP interception	TTL, hashing, attempt limit
Token replay	Single-use token bound to action
Privacy leakage	No default raw audio storage
Detector failure	Fail closed

OTP rules

6 digits
 60 seconds TTL
 Maximum 3 attempts
 Hashed storage
 Bound to action ID
 Single use
 Never logged in plaintext

Approval token rules

Random 256-bit token
 30-60 seconds TTL
 Single use
 Bound to action ID
 Hashed storage

K. Final data and evaluation plan

Languages

Minimum:
 Hindi
 Indian English
 Hinglish
 One regional language

Evaluation matrix

Test	Condition	Report
Baseline	Genuine vs synthetic	Detection rate, false-positive rate
Short audio	0.5, 1, 2, 4 seconds	Accuracy by duration, first alert latency
Language	Hindi, Indian English, Hinglish, regional	Separate results
Channel	Clean, telephone, codec, noise	Degradation
Genuine difficulty	Accent, noise, low volume	False alarms
Unseen generator	Held-out generator family	Detection at fixed false-positive rate
Partial manipulation	Synthetic segment inside genuine call	Alert timing
Workflow failure	Silence, timeout, stream loss, OTP expiry	Action remains unapproved

Metrics to show

Metric	Why it matters
Detection rate at fixed false-positive rate	Operational usefulness
False alerts per genuine call	User friction
First-alert latency median/p95	Real-time proof
Inference latency	Engineering proof
Blocked sensitive actions	Prevention proof
Expired OTP actions	Fail-safe proof
Stream failure actions	Robustness proof
Per-language results	Indian-language support

L. Final development roadmap

Phase 0: Before hackathon

Prepare:

- Repository.
- Docker Compose.
- Silero VAD running locally.
- One detector running locally.

- Test clips.
- Frontend skeleton.
- Dataset manifest.
- Fallback clips.
- Model latency check.

Phase 1: Hours 0-6

Goal:

Live audio stream produces real model scores.

Tasks:

- Session API.
- WebSocket audio.
- Silero VAD.
- Ring buffer.
- Detector inference.
- Raw score display.

Phase 2: Hours 6-12

Goal:

Stable risk scoring with insufficient-evidence handling.

Tasks:

- Quality checks.
- Evidence duration.
- Smoothing.
- Reason codes.
- Risk timeline.

Phase 3: Hours 12-18

Goal:

Sensitive action cannot complete without verification.

Tasks:

- Protected action API.
- Action state machine.
- OTP challenge.
- Approval token.
- Backend blocking.

This is the most important phase.

Phase 4: Hours 18-26

Goal:

Reproducible evaluation.

Tasks:

- Duration tests.
- Language tests.
- Channel tests.
- Genuine hard negatives.
- Synthetic clips.
- Result report.

Phase 5: Hours 26-32

Goal:

One screen explains the decision.

Tasks:

- Dashboard.
- Audit trail.
- Model/threshold version display.
- Fallback file mode.
- Demo controls.

Phase 6: Hours 32-36

Goal:

Repeatable demo.

Tasks:

- Regression tests.
 - OTP timeout test.
 - Stream disconnect test.
 - Detector unavailable test.
 - Docker packaging.
 - Rehearsal.
 - Q&A one-pager.
-

6. Final PPT blueprint

Use this slide structure.

Slide 1: Title

Title:

VIGILVOICE - Voice Trust Gate

Subtitle:

Real-Time Detection and Prevention of Voice-Cloning Impersonation Attacks

One line:

A voice-fraud verification layer that blocks sensitive actions until independent confirmation

Include:

- Team name
 - SIH26104
 - Problem statement
 - Organization
-

Slide 2: Problem

Heading:

Voice cloning can defeat voice-based trust

Bullets:

- Attackers can clone voices and request sensitive actions.
- Existing detectors fail under unseen generators, noise, codecs, replay, and Indian-language diversity.
- Detection alone is not enough.
- The system must act before the sensitive action completes.

Visual:

Cloned voice -> Call agent -> Sensitive action -> Fraud

Speaker line:

“The problem is not just detecting fake audio. The problem is stopping the action before it completes.”

Slide 3: Why existing solutions are insufficient

Heading:

Benchmark accuracy is not production safety

Bullets:

- Detectors can degrade on unseen generators.
- RTC codecs and telephone audio remove artifacts.
- Replay/re-recording can make fake audio look natural.
- Short audio creates uncertainty.
- Binary fake/real labels do not handle risk.

Visual:

Table:

Benchmark -> High accuracy

Real call -> Codec/noise/replay/unseen generator

Result -> Uncertainty

Need -> Action policy

Speaker line:

“We assume detectors are imperfect. The system must still prevent unsafe actions.”

Slide 4: Our solution thesis

Heading:

Detection buys evidence. Prevention requires action control.

Core statement:

We convert voice-risk evidence into an authorization decision before a sensitive action comp

Bullets:

- Continuous voice analysis.
- Dynamic risk scoring.
- Step-up verification.
- Action blocking.
- Audit trail.

Visual:

Voice evidence -> Risk state -> Action policy -> Independent verification -> Safe approval/t

Slide 5: Demo scenario

Heading:

A protected banking action during a live call

Scenario:

Caller requests transfer of INR 25,000.

Agent initiates protected action.

System analyzes voice.

Risk rises.

Action is blocked.

OTP verification is required.

If verification fails or expires, action remains unapproved.

Visual:

Call flow diagram.

Speaker line:

“This is not a detection dashboard. It is an authorization control.”

Slide 6: System architecture

Heading:

End-to-end prevention pipeline

Use this diagram:

Browser / Demo Client

v

WebSocket PCM Stream

v

Silero VAD + Quality Gate

v

Evidence Window Builder

v

One Spoof Detector

v

Temporal Smoothing

v

Policy Engine

v

Risk State + Reason Codes

v

Protected Action Control

v

Independent OTP Verification

v

Approve / Block / Review

v
Audit Log

Speaker line:

“Each stage produces evidence. The policy engine decides whether the action can proceed.”

Slide 7: AI pipeline

Heading:

One detector, calibrated and temporally smoothed

Bullets:

- Silero VAD for speech segmentation.
- Quality gate for poor/insufficient audio.
- One spoof detector.
- 2-second evidence window by default.
- Duration study: 0.5s, 1s, 2s, 4s.
- Temporal smoothing avoids unstable decisions.
- Reason codes explain every state.

Speaker line:

“We deliberately avoid model overload. One detector earns its place before additional branches are added.”

Slide 8: Risk states

Heading:

Risk is evidence, not identity

Table:

State	Meaning
LOW	No strong synthetic-speech evidence detected
ELEVATED	Elevated synthetic-speech indicators
HIGH	Strong synthetic-speech indicators
INSUFFICIENT_EVIDENCE	Not enough voiced evidence
POOR_QUALITY	Audio quality too low
SERVICE_UNAVAILABLE	Fail closed

Important line:

LOW risk does not mean the caller is authenticated.

Slide 9: Prevention policy engine

Heading:

The system fails closed under uncertainty

Table:

Condition	Action
LOW	Continue monitoring; app authentication still applies
ELEVATED	Step-up verification
HIGH	Block pending review
INSUFFICIENT	Verification/review
POOR QUALITY	Verification/review
DETECTOR FAILURE	Verification/review
OTP EXPIRED	Action remains unapproved

Speaker line:

“The most important security property is that uncertainty never becomes silent approval.”

Slide 10: Independent verification

Heading:

Voice risk is never the only trust anchor

Bullets:

- OTP/push/email is independent from the voice channel.
- Challenge is bound to the action.
- Token is single-use.
- TTL is short.
- Attempts are limited.
- Backend enforces verification.

Speaker line:

“Even if the voice sounds convincing, the action still requires independent confirmation when risk is elevated.”

Slide 11: Indian-language and data plan

Heading:

Built for Indian-language call conditions

Bullets:

- Hindi
- Indian English
- Hinglish
- One regional language
- Synthetic speech from IndicSynth
- Genuine speech from permitted datasets/team recordings
- Codec/noise/telephone conditions
- Speaker-disjoint and generator-disjoint evaluation

Speaker line:

“We do not treat Indian-language support as optional. It is part of the evaluation plan.”

Slide 12: Evaluation and proof

Heading:

We measure prevention, not just detection

Metrics:

- Detection rate at fixed false-positive rate.
- False alerts per genuine call.
- First-alert latency.
- Inference latency.
- Per-language performance.
- Per-duration performance.
- Channel degradation.
- Blocked sensitive actions.
- Expired OTP actions.
- Stream-failure actions.

Speaker line:

“We report performance by condition, not one misleading headline accuracy.”

Slide 13: Security, privacy, and audit

Heading:

Privacy by default and auditable decisions

Bullets:

- No raw audio storage by default.
- Short-lived audio buffers.
- Reason codes on every decision.
- Model and threshold versions logged.
- OTP hashed and short-lived.
- Approval tokens single-use.
- Fail-safe under detector failure.

Speaker line:

“The system produces evidence without retaining full conversations by default.”

Slide 14: Tech stack and deployment

Heading:

Lightweight, local, and demonstrable

Stack:

- FastAPI
- WebSocket
- Silero VAD
- PyTorch/ONNX Runtime
- PostgreSQL
- React/simple frontend
- Docker Compose
- Local fallback

Speaker line:

“The demo can run locally without depending on external APIs.”

Slide 15: Roadmap

Heading:

Start small, earn complexity

Phases:

Phase	Scope
Phase 0	Environment, data, model prep
Phase 1	Streaming + VAD + detector
Phase 2	Quality + temporal smoothing
Phase 3	Action blocking + OTP
Phase 4	Evaluation matrix
Phase 5	Dashboard + audit
Phase 6	Regression + rehearsal

Future:

- Speaker verification.
- Replay branch.
- OOD branch.
- Few-shot adaptation.
- WebRTC/telephony adapters.

Speaker line:

“Additional branches are earned through validation, not added blindly.”

Slide 16: Differentiation

Heading:

Why this is different

Use four points:

1. **Prevention-first**
 - Blocks sensitive action before completion.
2. **Uncertainty-aware**
 - Handles insufficient evidence and poor audio.
3. **Indian-language aware**
 - Evaluates Hindi, Indian English, Hinglish, and one regional language.
4. **Auditable**
 - Every decision has reason codes, model version, and threshold version.

Speaker line:

“Our innovation is not one model. It is the evidence-to-action workflow.”

Slide 17: Live demo plan

Heading:

Six demonstration cases

Case	Expected outcome
Genuine call	Low evidence, normal flow
Cloned call	Elevated/high risk, action blocked/step-up
Telephony/noisy clone	Risk with channel reason codes
Difficult genuine speech	No false certainty; step-up if insufficient
Unseen generator	Uncertainty handled safely
System failure	Action remains unapproved

Speaker line:

“We show success and failure modes openly.”

Slide 18: Limitations and responsible deployment

Heading:

We do not overclaim

Bullets:

- No detector can catch all synthetic voices.
- LOW risk does not prove identity.
- Voice risk is supplementary to authentication.
- Uncertain cases require verification or review.
- Additional branches are future work.

Speaker line:

“The system is designed to be safe when it is uncertain.”

Slide 19: Conclusion

Heading:

From detection to prevention

Final statement:

VIGILVOICE converts voice-risk evidence into a controlled authorization decision, preventing

7. Final demo script

Use this exact flow.

Demo 1: Genuine call

Input:

- Genuine Hindi or Indian English speech.

UI shows:

Risk State: No strong synthetic-speech evidence detected

Evidence: 2.4 seconds voiced

Action: Fund transfer requested

Verification: OTP required for high-value action

Outcome: Approved after OTP

Judge takeaway:

Low-risk genuine speech does not create unnecessary friction.

Demo 2: Cloned call

Input:

- Synthetic cloned voice.

UI shows:

Risk State: Elevated synthetic-speech indicators

Reason: SYNTH_SCORE_ELEVATED

Action: Fund transfer requested

Verification: Required

OTP: Not verified/expired

Outcome: Blocked

Judge takeaway:

Detection leads to prevention.

Demo 3: Telephony/noisy clone

Input:

- Clone with codec/noise.

UI shows:

Risk State: Elevated/High

Reasons: SYNTH_SCORE_ELEVATED, NARROWBAND_CHANNEL

Outcome: Step-up or block

Judge takeaway:

The system considers channel difficulty.

Demo 4: Difficult genuine speech

Input:

- Genuine speaker with noise/accent/low volume.

UI shows:

Risk State: Poor quality or insufficient evidence

Outcome: Step-up verification required

Judge takeaway:

The system knows when it does not have enough evidence.

Demo 5: Unseen generator

Input:

- Synthetic voice from unseen generator.

UI shows:

Risk State: Elevated/High or uncertain

Outcome: Step-up or review

Judge takeaway:

The system is designed for unknown attacks.

Demo 6: System failure

Input:

- Detector stopped or stream disconnected.

UI shows:

Risk State: Service unavailable/stream interrupted

Outcome: Sensitive action remains unapproved

Judge takeaway:

The system fails closed.

8. Final proof metrics for PPT

Show a small metrics panel:

First alert latency median: X.X s
First alert latency p95: X.X s
Detection rate at fixed FPR: XX%
False alerts per genuine call: X.XX
Blocked sensitive actions: 4/4
Expired OTP actions blocked: 1/1
Stream failure actions blocked: 1/1

Do not fake numbers. Generate them from your evaluation runner.

9. Judge Q&A preparation

Q1: How is this different from a normal deepfake detector?

Answer:

A normal detector only gives a score. Our system converts the score into an authorization decision. If the voice evidence is risky, uncertain, or unavailable, the sensitive action cannot complete without independent verification.

Q2: Why are you not using many models?

Answer:

We intentionally use one detector in the critical path. Additional branches such as speaker verification, replay detection, and OOD scoring are added only if evaluation shows they improve prevention enough to justify latency and complexity.

Q3: What if the detector is wrong?

Answer:

The system is designed to fail safely. Poor audio, insufficient evidence, detector failure, or expired verification all keep the sensitive action unapproved.

Q4: How do you handle Indian languages?

Answer:

We evaluate Hindi, Indian English, Hinglish, and one regional language. We separate genuine speech, synthetic speech, code-switching, and channel conditions in the test matrix.

Q5: How do you prove real-time?

Answer:

We report first-alert latency, inference latency, evidence duration, stream stability, and hardware used.

Q6: Why not just use speaker verification?

Answer:

Speaker verification checks identity similarity, but a cloned voice can still match the target speaker. Spoof evidence and independent verification remain necessary.

Q7: What about replay attacks?

Answer:

Replay is treated as a test condition. For MVP, channel/quality signals and fail-safe policy handle uncertainty. A dedicated replay branch is a Phase-2 addition if evaluation justifies it.

Q8: Is this privacy-safe?

Answer:

The prototype avoids raw audio retention by default. It stores scores, reason codes, action events, model version, and threshold version. Debug audio storage is optional and short-lived.

Q9: Can attackers adapt?

Answer:

Yes. That is why we use generator-disjoint evaluation, uncertainty handling, and a roadmap for adaptation. We do not claim perfect detection. We claim safer authorization behavior.

Q10: What actually prevents fraud?

Answer:

The protected action remains pending until valid independent verification succeeds. This backend enforcement is the prevention layer.

10. Final do/don't list for the PPT

Do

- Show prevention before action.
- Show reason codes.
- Show uncertainty handling.
- Show fail-closed behavior.
- Show Indian-language plan.
- Show latency metrics.
- Show local fallback.
- Show limitations honestly.
- Show one clean architecture diagram.
- Show backend enforcement.

Don't

- Don't claim 100% detection.
- Don't say "trusted caller."
- Don't make watermarking the core.
- Don't build many models in MVP.
- Don't rely only on EER.
- Don't hide failures.

- Don't depend on internet APIs during demo.
 - Don't claim legal compliance.
 - Don't say voice risk proves identity.
 - Don't overbuild during the 36-hour sprint.
-

11. Final one-paragraph pitch

Use this in your PPT introduction:

Voice-cloning attacks can compromise voice-based workflows when a cloned voice requests a sensitive action. Existing detectors often degrade under unseen generators, codecs, noise, replay, and short audio. VIGILVOICE addresses this by treating detection as evidence, not as final truth. The system continuously analyzes call audio using Silero VAD, a quality gate, one calibrated spoof detector, and temporal smoothing. It converts voice evidence into a risk state and blocks or pauses sensitive actions until independent verification succeeds. Under poor audio, insufficient evidence, or detector failure, the system fails closed. This makes the prototype not just a detector, but a prevention layer for real-time voice interactions.

12. Final build priority order

If you are starting now, build in this exact order:

1. Repository and Docker Compose.
 2. FastAPI backend with session API.
 3. WebSocket PCM streaming.
 4. Silero VAD integration.
 5. Quality gate and evidence window builder.
 6. One spoof detector inference.
 7. Risk state and reason codes.
 8. Protected action API.
 9. OTP verification service.
 10. Backend action blocking.
 11. One-screen dashboard.
 12. Audit log.
 13. Evaluation runner.
 14. Fallback clips.
 15. Rehearsal.
-

13. Final winning narrative

Your final narrative should be:

We are not building another fake-voice classifier. We are building a voice-trust gateway that assumes detectors can fail, measures uncertainty honestly, and prevents sensitive actions from completing unless the evidence and independent verification allow it.

That is the strongest message across all three documents.